

YUEZHEN (LILY) DONG

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RELEVANT SKILLS

Languages	Python, Java, C/C++, TypeScript/JavaScript, SQL, HTML/CSS, Ruby, Bash
Frameworks	Spring Boot, React/Next.js, Node/Express, Three.js, Flask, Pygame, Arduino, GraphQL
Developer Tools	AWS, Docker, Kubernetes, Git, Maven, Linux, PostgreSQL, AutoCAD, VSCode, IntelliJ

EDUCATION

University of Waterloo - Faculty of Engineering	Expected Graduation MAY 2029
BASc in Electrical Engineering	Waterloo, ON, CAN

ACHIEVEMENTS

IBDP Graduate CaYPT (Canadian Young Physicists' Tournament) National Champion IYPT (International Young Physicists' Tournament) Bronze Medalist FIRST Robotics Provincial Semi-Finalist

WORK EXPERIENCE

Software Engineering Intern	SEP 2025 – DEC 2025
React, TypeScript, Ruby, GraphQL	Shopify

- Built activation/onboarding flows that guided new retail merchants from install to first in-person sale (guided setup, device login, POS entry points), implementing **React + TypeScript** UIs backed by **Ruby/GraphQL** services; reduced average onboarding time by **35%** and improved day-0 adoption for **500+ merchants**.

Web Developer Coop	JAN 2025 – APR 2025
Java, Spring Boot, Maven, Apache POI, Git	AGF Investments

- (In production — company-wide) Developed a **Java-based** ScoreCard application with robust file archiving, large XLSX handling via **Apache POI streaming**, data validation/transformation, and a flexible properties-based configuration system; now used in **daily reporting workflows across 4 departments**, saving **10+ staff hours/week**.
- (In production — company-wide) Delivered an end-to-end **Spring Boot** reporting pipeline to ingest XLSX, filter by date/time, archive inputs, and generate Excel/CSV outputs; automated email distribution under **Maven/Git**, eliminating **100%** of manual reporting steps and reducing delivery time from hours to minutes.
- Built **Pega** scenario tests and **Selenium** UI/service tests covering 90% of core business flows, reducing regression defects by **40%** and enabling releases to ship **2x faster**.

Website Developer	DEC 2020 – DEC 2023
Java, HTML/CSS, JavaScript	Beyond the Walls / Kindness Catalogue

- Designed and deployed a dynamic website using **Java** backend and **HTML/CSS, JavaScript** frontend, ensuring mobile responsiveness and smooth interaction; maintained uptime of **99.9%** and supported **5,000+ monthly visitors**.

PERSONAL PROJECTS

Ecoland: AI Ecosystem Simulator
Java, JavaFX, OOP, AI, GeneticAlgorithms

- Built a **Java** simulator with custom **AI algorithms**, pathfinding, and genetic algorithms to model various species interactions.
- Employed **concurrency**, **data structures**, and **OOP** principles for large-scale simulations, enabling real-time monitoring, performance tuning, and modular feature expansions.

3D Data Sandbox
Electron, Node.js, Three.js, PapaParse, JavaScript

- Built a cross-platform **Electron** desktop application using **Three.js** and **Node.js/PapaParse** for offline 3D visualization, parsing, and analysis of local CSV/JSON datasets.
- Implemented scatter, grid, and K-means layouts with outlier detection, correlation lines, and user-friendly UI controls for filtering, highlighting, and exporting.

Puppy Portal Puzzle: Time-Trotting Dogventure
Python, Pygame, Pathfinding, Time-Manipulation, Puzzle Architecture

- Developed a controlled **time-travel** mechanic with simultaneous clones, integrating **BFS pathfinding** for AI enemies.
- Structured the game using **Python OOP** and event-driven architecture, implementing multi-layer switch logic, portal systems, and crate interactions while ensuring seamless performance under increasing complexity.